

Worksheet 7C: Eigenvalues and Eigenvectors — Solutions

Series 7: Linear Algebra
AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Instructor Copy — Not for Distribution

Note to instructor: Key answers are given. Accept mathematically equivalent forms.

Part A

- (a) $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$; $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. (b) Yes; eigenvalues 3 and 2. (c) $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \neq \lambda \begin{bmatrix} 1 \\ 1 \end{bmatrix}$: not an eigenvector. (d) A stretches the x -axis by 3 and the y -axis by 2.
- $A = 5I$. Every non-zero vector is an eigenvector. Infinitely many.
- (a) No real vector stays pointing in the same direction after a 90° rotation (it rotates to a perpendicular direction). (b) No real eigenvalues.
- (a) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (eigenvalue 1) and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (eigenvalue -1). (b) Eigenvalue 1: vector unchanged (lies on reflection axis). Eigenvalue -1 : vector reversed (flipped through the axis).
- $A(2\mathbf{v}) = 2A\mathbf{v} = 2\lambda\mathbf{v} = \lambda(2\mathbf{v})$. ✓
- $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \lambda\mathbf{v} + \lambda\mathbf{w} = \lambda(\mathbf{v} + \mathbf{w})$. ✓
- From $A\mathbf{v} = \lambda\mathbf{v}$: $\mathbf{v} = A^{-1}(\lambda\mathbf{v}) = \lambda A^{-1}\mathbf{v}$, so $A^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$.
- I : eigenvalue 1 for every vector. O : eigenvalue 0 for every vector.

Part B

- $\lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$; $\lambda = 5, 2$.
- $\lambda^2 - 8\lambda + 17 = 0$; $\lambda = (8 \pm \sqrt{64 - 68})/2$: complex eigenvalues $(4 \pm i)$.
- $\lambda^2 - 5\lambda + 4 = (\lambda - 4)(\lambda - 1) = 0$; $\lambda = 4, 1$.
- $\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2 = 0$; $\lambda = 2$ (repeated).
- $\lambda^2 + 1 = 0$; $\lambda = \pm i$ (complex).
- Problem 9: $\text{tr} = 7 = 5 + 2\checkmark$; $\det = 10 = 5 \times 2\checkmark$. Problem 11: $\text{tr} = 5 = 4 + 1\checkmark$; $\det = 4 = 4 \times 1\checkmark$.
- $\lambda_1\lambda_2 = \det(A) = 0$, so at least one eigenvalue is 0. A is not invertible ($\det A = 0$).

16. Diagonal entries: $\lambda = 2, 3, -1$.
17. Exactly 3 (counting multiplicity, including complex).
18. $\lambda^2 - 5\lambda + 5 = 0$; $\lambda = (5 \pm \sqrt{5})/2 \approx 3.618, 1.382$. Both positive — yes, positive definite.

Part C

19. $\lambda = 5$: $(A - 5I) = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$; $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. $\lambda = 2$: $(A - 2I) = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $\mathbf{v} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$.
20. $\lambda = 4$: $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. $\lambda = 1$: $\mathbf{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.
21. Standard basis vectors: $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ ($\lambda = 3$) and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ ($\lambda = 5$).
22. $(D - 2I) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$; only $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. One independent eigenvector — cannot span $\mathbb{R}^2$.
23. $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \checkmark$; $A \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -2 \end{bmatrix} \checkmark$.
24. $A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\lambda = 4$. Other eigenvalue: $\text{tr} = 6 \Rightarrow \lambda_2 = 2$; eigenvector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.
25. $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 - 1 = 0$. $\checkmark$ Orthogonal.
26. $\lambda^2 - 1.3\lambda + 0.3 = 0$; $\lambda = 1$ and $\lambda = 0.3$. For $\lambda = 1$: $(P - I)\mathbf{v} = \mathbf{0}$; $\mathbf{v} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ (proportional to $[4/7, 3/7]$). This matches the stationary distribution from ws07d.
27. $\lambda^2 - 7\lambda + 8 = 0$; $\lambda = (7 \pm \sqrt{17})/2 \approx 5.56, 1.44$. First PC (largest $\lambda \approx 5.56$): eigenvector approximately $\begin{bmatrix} 0.82 \\ 0.57 \end{bmatrix}$.
28. $5.56/(5.56 + 1.44) = 5.56/7 \approx 79.4\%$ of variance.

Part D

29. $P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$, $\Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$. $P\Lambda P^{-1} = A$. $\checkmark$
30. $A^3 = P\Lambda^3 P^{-1}$; $\Lambda^3 = \begin{bmatrix} 125 & 0 \\ 0 & 8 \end{bmatrix}$; compute product.
31. $\lambda = 1$ (eigenvector $[4, 3]^T \propto$ steady state) and $\lambda = 0.3$. As $n \rightarrow \infty$: $(0.3)^n \rightarrow 0$; each row converges to the stationary distribution $[4/7, 3/7]$.
32. Only one independent eigenvector spans a 1D subspace; you cannot form an invertible P from a 2×2 matrix with only one independent eigenvector.

33. Covariance: $\sigma_{ij} = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \sigma_{ji}$, so $\Sigma^T = \Sigma$. The Spectral Theorem guarantees real eigenvalues, orthogonal eigenvectors, and full diagonalisability — making PCA well-defined.
34. (a) $3\lambda_1, 3\lambda_2$ (b) $\lambda_1 + 5, \lambda_2 + 5$ (c) λ_1^2, λ_2^2 (d) $1/\lambda_1, 1/\lambda_2$
35. (a) $\text{tr} = 2 + (-1) + 3 = 4$ (b) $\det = 2 \times (-1) \times 3 = -6$ (c) $\det \neq 0$ — invertible.
36. (a) Reduces storage/computation from 1000 dimensions to 10. (b) 5% of variance information is lost. (c) Rank of the projected data is at most 10 — we project onto a rank-10 subspace.

Part E

37. $\pi P = \pi \Rightarrow P^T \pi^T = \pi^T$: π^T is a right eigenvector of P^T with eigenvalue 1.
38. The damped matrix is ergodic (all states accessible), ensuring by the Perron–Frobenius theorem that $\lambda = 1$ has multiplicity 1 and a unique normalised eigenvector. Without uniqueness, the ranking would be arbitrary.
39. $A^T A = \begin{bmatrix} 11 & 7 & 7 \\ 7 & 11 & 7 \\ 7 & 7 & 11 \end{bmatrix}$. Eigenvalues: $\lambda_1 = 25, \lambda_2 = \lambda_3 = 4$. Singular values: 5, 2, 2.
40. (a) Together they answer: what can A compute ($\times$), what does A solve ($A\mathbf{x} = \mathbf{b}$), and what is A 's intrinsic geometry (eigenvectors). (b) $\mathbf{x}^T A \mathbf{x}$ is maximised along the eigenvector with the largest eigenvalue — that direction captures the most “energy”, hence the most variance.